ultimate destruction of all things. At some point, in a cosmic sense, it will not have mattered that we ever lived. The universe will, more likely than not, fade into a cold, dark, empty cosmos, and all that we've done will be utterly forgotten. Where does that leave us now?

Hiranya Peiris sums it up in one word: "sad."

"It's very depressing," she says. "I don't know what else to say about it. I give talks where I mention that this is probably the fate of the universe, and people have cried."

It does provoke some perspective-taking. "It's very intriguing to me that the universe has produced a very interesting period where a lot happens," she says. "And yet we seem to face a much longer period in utter darkness, cold. It's horrible. I feel, actually, from that point of view, very lucky, to be in the few years of cosmology where we are learning all this stuff for the first time."

"It momentarily makes me feel sad," agrees Andrew Pontzen. "Then I very quickly start worrying about our problems here on Earth right now and think, 'Come on.' We're in so much deeper trouble than the Heat Death of the universe. So I suppose it makes me start thinking about the problems that we face as a civilization on much shorter timescales. If I'm going to worry about anything, it's gonna be those, not the Heat Death."

"I just don't really have an emotional connection, I suppose, to the death of the universe," Pontzen continues, "but I do to the death of the Earth. I don't mind the fact that I'm going to die in fifty years or whatever, but I don't want the Earth to die in fifty years."

I have a lot of sympathy for this view. In terms of things we should actually worry about, the Heat Death, or vacuum decay, or the Big Rip, or whatever, cannot be at the top of that list (even setting aside the fact that we are utterly powerless to do anything about them). As living beings, we naturally care most about our own lives, and the lives of those close to us in space and time, and for the most part we leave the unimaginably distant cosmic future to its own devices.

But personally, I still feel there's a big difference, in some emotional sense, between "we go on forever" and "we don't." Nima Arkani-Hamed feels the same way. "At the absolute, absolute deepest level … whether or not people explicitly admit to thinking about it or not (and if they don't they're all the poorer for it) … If you think there is a purpose to life, then I at least don't know how to find one that doesn't connect to something that